Rajan Vivek

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Education

Stanford University 2024
MS Computer Science (GPA: 4.06 / 4.00) *Palo Alto, CA*
- **Courses:** Machine Learning, NLP, Foundation Models, Speech Processing, Statistics, Decision Making w/ Uncertainty

Georgia Tech 2022
BS Electrical Engineering (GPA: 3.97/4.00) *Atlanta, GA*
- **Courses:** Statistical ML, Cloud Computing, Python, C++, Signal Processing, Data Structures & Algorithms

Experience

Contextual AI 2024-Present
Member of Technical Staff (Research) *Mountain View, CA*
- Owned real-time hallucination detection feature in company's flagship product end-to-end with 83% F1-score. Required synthetic data generation+eval, reward model fine-tuning, full-stack implementation in production
- Led LLM post-training for flagship Grounded Language Model achieving #1 on the leading groundedness benchmark (FACTS) (outperforming Gemini 2.5 Pro, Opus 4, GPT-5) + best open-weights model on customer evals
- Co-Led training of SoTA reward model (LMUnit), achieving #1 on Reward Bench 2 + FLASK + BigGenBench (as of June 2025) using synthetic data and multi-loss strategy. Led to first-author EMNLP 2025 publication.

Scale AI Summer 2023
Machine Learning Research Intern *San Francisco, CA*
- Designed spatio-temporal Q-former architecture for video/vision language model through extensive architecture and data ablations, achieving performance surpassing contemporary SoTA models (InstructBLIP, VideoLLaMA, & MPlugOwl) on NextQA video question answering

Stanford NLP Group 2022-2024
Researcher *Palo Alto, CA*
- Designed novel representative data selection technique for efficient language model evaluation, allowing rapid discovery of model weaknesses across a large dataset. Led to first author EACL 2024 publication.
- Ran many experiments exploring research ideas including benchmark hill-climbing during LM pretraining, data-relatedness metrics, zero-shot vs. fine-tuned performance, meta-learning for synthetic training data generation

JP Morgan Chase Summer 2022
AI & Data Science Analyst *Jersey City, NJ*
- Trained and productionized a transformer-based entity recognition model for invoice processing. Designed custom data augmentation techniques, improving model performance by 11% (F1-score). AWS Lambda, S3, DynamoDB.

Publications

LMUnit: Fine-Grained Evaluation with Natural Language Unit Tests | *EMNLP Findings 2025*
- **Rajan Vivek***, William Berrios*, Jon Saad-Falcon*, Nandita Naik, Douwe Kiela, Shikib Mehri, et al. (*co-first author)
- SoTA reward model (#1 on RewardBench 2 & more) + proposed LLM evaluation paradigm backed by human studies

Anchor Points: Benchmarking Models with Much Fewer Examples | *EACL Main 2024*
- **Rajan Vivek**, Kawin Ethayarajh, Diyi Yang, Douwe Kiela
- Novel technique to select data subset for highly efficient eval + elucidate model behavior across the entire dataset

Explainable Activity Recognition for Smart Home Systems | *ACM Transactions on Interactive Intelligence 2023*
- Devleena Das, Yasutaka Nishimura, **Rajan Vivek**, Naoto Takeda, Sean Fish, Thomas Plotx, Sonia Chernova
- Framework that applies explainable AI methods (LIME, SHAP) to sensor time series, improving caregiver monitoring

Technical Skills

Foundations: PyTorch, Transformers, Python, distributed training (DDP, FSDP, DeepSpeed ZeRO), experiment tracking
Specialization: RLHF, reward modeling, SFT, DPO, GRPO, synthetic data pipelines, evaluation, agentic systems

Awards

NSF Graduate Research Fellowship Honorable Mention (2022)
Outstanding Project Award, Stanford CS 330: Deep Multi-Task & Meta Learning (2022)
Two 1st place awards, 2nd overall (out of 250 teams), HackGT (2019)